Structural constraint of generative output in confidentiality-bearing legal documentation: A local audio-to-notes system and its design principles

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ABSTRACT

Generative language models now produce fluent summaries of spoken proceedings, but in legal work fluency is not the binding constraint: a summary that invents a holding is worse than none, and audio of a privileged proceeding often cannot lawfully be sent to a third-party service at all. Local tools address the second problem and inherit the first. We report the design and technical evaluation of Lausanne, an on-device system converting legal proceeding audio into speaker-attributed, citation-backed notes on consumer hardware. Its central move is to constrain *where output text originates* rather than instruct the model about what to write: in one mode the response schema exposed to the model contains only integer identifiers of source utterances and no free-text field for claims, enforced at decode time, so note text is re-materialised by lookup and fabrication is inexpressible rather than improbable. We further show that confidentiality claims must audit model identity and not only endpoint address, reporting a failure class in which a cloud-hosted model reached through a loopback URL satisfies a locality check while transmitting the entire transcript. We formalise eleven design principles and report a criticality-disaggregated word error rate substantially higher on legally load-bearing tokens than in aggregate, identifying an error surface upstream of the hallucination literature: extraction guarantees faithfulness to the transcript, not to the proceeding.

1. Introduction

Legal work is unusually language-intensive, and much of it consists of converting speech into a written record: hearings, depositions, client intake, oral argument, internal case discussion. That work is slow, repetitive from the practitioner's point of view, and therefore an obvious candidate for automation with contemporary generative models.

It is also work in which two constraints bind simultaneously, and the literature has largely treated them separately. The first is reliability. Dahl, Magesh, Suzgun, and Ho (2024) document hallucination rates of 58–88% on specific, verifiable queries about federal cases, and show that models frequently fail to flag their own errors; Magesh et al. (2025) find that even retrieval-augmented commercial legal research tools hallucinate between roughly 17% and 33% of the time. The second is confidentiality. Audio of a privileged proceeding is categorically restricted: disclosure to a third party is often impermissible regardless of that party's security posture, because the obligation attaches to disclosure itself rather than to the risk of breach.

Local deployment answers the second constraint and inherits the first. A system that runs a language model on the practitioner's own machine transmits nothing, and still writes whatever the model produces. The prevailing response is to instruct the model not to fabricate, and then to measure how often it does anyway.

This paper takes a different route. We observe that faithfulness is a property of *where output text originates*, not of *what selects it*, and that these can be separated. In the system reported here, one operating mode gives selection to a language model while restricting its entire output surface to integer identifiers drawn from a numbered transcript. The response schema contains no free-text field for claims, and the constraint is enforced at decode time by the inference engine's grammar. Note text is then reconstructed from the transcript by identifier lookup. Selection is as good as the model's judgement; wording cannot be anything other than source.

We make three contributions. First, we describe and formalise this constraint mechanism, and distinguish it from negative prompting, which steers a model probabilistically rather than removing the channel. Second, we report a confidentiality failure mode that endpoint-level auditing does not catch, in which a cloud-hosted model reached through a loopback address would pass a locality check while transmitting the whole transcript; we argue that provenance claims must audit model identity as well as address. Third, we formalise eleven design principles in the template of Chandra, Seidel, and Gregor (2015), and report an evaluation design in which word error rate is disaggregated by whether a token is legally load-bearing.

Section 2 reviews the relevant literature and locates the contribution among other ways of bounding generative scope. Section 3 describes the artefact. Section 4 states the design principles. Section 5 reports the evaluation. Section 6 discusses two findings we regard as generalisable, and Section 7 states limitations, of which the most important is that no legal practitioner has yet used the system.

2. Background

2.1. Reliability and hallucination in legal settings

The reliability problem in legal AI is acute and well documented. Beyond the rates already cited, Dahl et al. (2024) note that models tend to accept a user's incorrect legal premises rather than correct them, and that the users most likely to rely on such tools — self-represented litigants — are precisely those least able to detect the errors. The professional consequences are not hypothetical: sanctions following the citation of fabricated cases have made legal caution toward generative systems a rational risk assessment rather than an aversion to be designed around.

Because human evaluation of open-ended legal generation is expensive, the field increasingly evaluates with models. That paradigm carries its own reliability questions. Pointwise scoring is

poorly calibrated and prompt-sensitive; checklist decomposition improves interpretability and reduces susceptibility to stylistic cues (Lee et al., 2024); and, structurally distinct from ordinary generation, the text being evaluated is an untrusted input that can attempt to hijack the evaluator. An unreliable generator assessed by an exploitable judge yields a benchmark that looks healthy while concealing failure.

Two further findings bear on any evaluation that delegates judgement to a model. Park, Greenewald, Alim, Wang, and Azizan (2025) show that reward models used to score intermediate reasoning are poorly calibrated, and argue for quantifying a model's uncertainty about its own verdicts rather than consuming those verdicts as point estimates. And Bashardoust, Feng, Shrestha, and colleagues (2026) demonstrate that language models allocating tasks exhibit systematic bias tied to identity labels attached to the material, which is to say that what surrounds the content being judged can move the judgement independently of the content itself.

We take a conservative reading of this literature. Where a property can be decided without a model, deciding it without a model removes calibration error, bias, and injection susceptibility together rather than mitigating each in turn. Section 5.1 describes the measure we use for that purpose, and Section 3.3 the design property that makes such a measure available at all.

2.2. Confidentiality as a design constraint

Work on creativity support tools built over foundation models has largely assumed that model hosting is an implementation detail. Herath, Sanoubar, Krishna, and Shrestha (2026), developing design principles for text-to-image creativity support tools in a commercial game studio, treat interface simplification and prompt structure as the principal design surfaces, and confidentiality does not appear among their boundary conditions. That is appropriate for their setting: concept art is commercially sensitive, not privileged.

Where material is privileged, the topology of inference becomes a first-class property of the artefact, and one the user must be able to demonstrate rather than merely be assured of. We take this as the point of departure for the present work, and return in Section 6.2 to a specific way in which demonstrating it is harder than it appears.

2.3. Bounding the scope of generative action

A useful way to locate this paper's contribution is to ask what a designer bounds when a generative model is powerful but not trustworthy. The literature on artificial-intelligence-augmented decision making (Shrestha, Ben-Menahem, & von Krogh, 2019; Raisch & Krakowski, 2021) offers several answers, and they are not alternatives so much as different axes.

One may bound *how much* the model decides. Shrestha et al. (2019) distinguish full human decision making, hybrid arrangements, and full delegation, and treat the choice among them as an organisational design question. One may bound *when* the model acts: Tripathi, Sharma, Mishra, Kumar, Herath, and Shrestha (2026), designing an LLM-augmented resilience analyst, invoke generative reasoning only where statistical detection is likely to fail, and describe this explicitly as preserving analytical rigour by confining the generative component to the part of the problem that needs it. One may instead *adapt* the model, as in the self-editing approach of Zweiger, Pari, Guo, Akyürek, Kim, and Agrawal (2025), where the model generates its own finetuning directives.

The axis this paper occupies is a fourth: bounding *what the model may emit*. Rather than restricting the model's authority, its moment of involvement, or its weights, we restrict the vocabulary of its response. The distinction matters because the other three bounds are enforced around the model — by an orchestrator, a trigger condition, or a training loop — whereas this one is enforced inside decoding, and therefore holds without the surrounding system having to be correct. It is also, unlike adaptation, free: it requires no additional training and no reward signal.

We regard these as complementary. A system may well bound scope on several axes at once, and Section 4 states principles that do so.

2.4. Design principles as prescriptive knowledge

We formalise our design knowledge using the template of Chandra et al. (2015), in which a principle names a material property of the artefact, the action potential that property affords its users, and the boundary conditions under which the design operates. This follows established practice in design science research (Gregor, Chandra Kruse, & Seidel, 2020) and permits reusability assessment (Iivari, Hansen, & Haj-Bolouri, 2021).

We are explicit that this is design science research and not action design research (Sein, Henfridsson, Purao, Rossi, & Lindgren, 2011). Action design research requires reciprocal shaping between artefact and organisational context, and its evidence comes from practitioners using the artefact in their own setting. No legal practitioner has used this system. Section 7 states what that costs and what would close it.

3. The artefact

Lausanne converts audio of a legal proceeding into speaker-attributed, time-stamped, citation-backed notes. Every model executes on the user's own machine. The reference configuration is a consumer laptop with an 8 GB GPU and 16 GB of system memory — chosen deliberately, since a confidentiality guarantee that requires a data-centre GPU is a weaker guarantee than one that runs on hardware a solo practitioner already owns.

3.1. Staged execution and the resource budget

The system comprises five stages (Fig. 1). Each executes as an isolated operating-system process communicating only through schema-validated JSON persisted to disk.

This is not architectural preference. The three model-bearing stages require approximately 12 GB of accelerator memory in combination against roughly 7.3 GB usable after the display reservation, so they cannot be co-resident. Releasing that memory is harder than it appears: common deep-learning runtimes do not return accelerator memory within a process, the accelerator context persists until the process exits, and local inference servers retain models between calls by default. Process termination is the only eviction primitive that reliably succeeds. Two further properties follow at no cost: recovery, since every intermediate is durable and a late failure does not invalidate earlier computation; and independent testability, since each stage is a function from one validated artefact to another and can be exercised with synthetic input on hardware with no accelerator at all.

A further consequence deserves emphasis for practitioners. On the Windows platform, exceeding accelerator memory does not raise an error; the driver spills to shared system memory and the pipeline completes correctly but roughly fifty times slower. The failure being guarded against is therefore not a crash but a silent degradation, which a user has no obvious means of diagnosing.

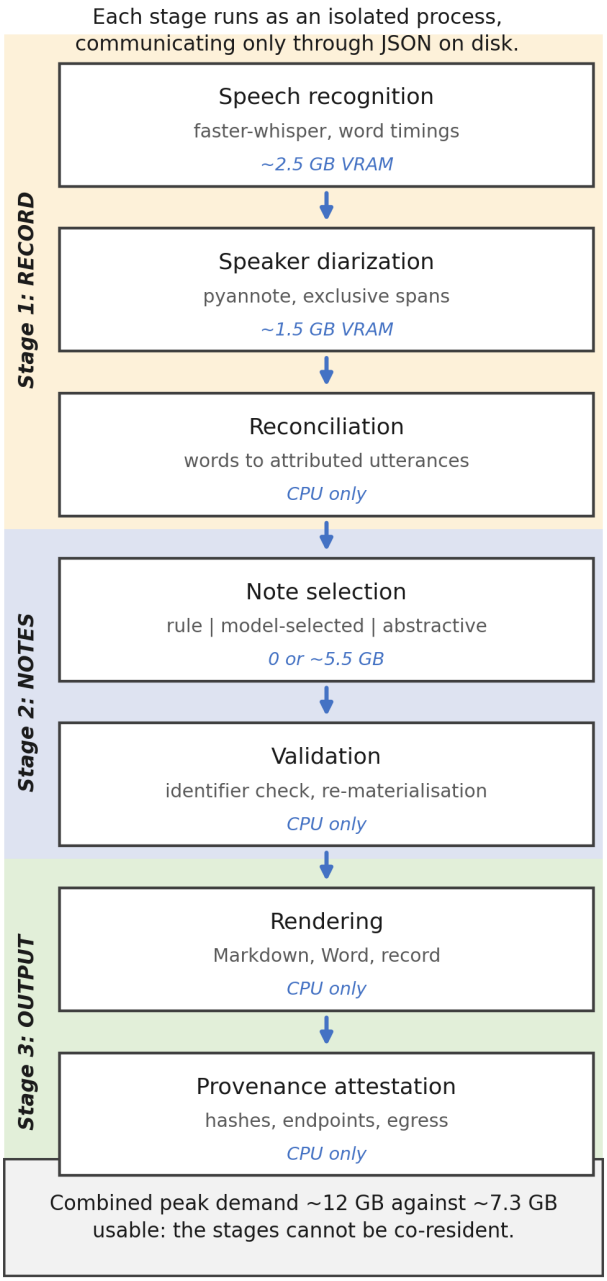


Fig. 1. Staged execution of the pipeline. Each stage runs as a separate operating-system process and communicates only through schema-validated JSON on disk. Combined peak demand exceeds available accelerator memory, which is why sequencing rather than co-residence is required.

3.2. Three modes of note generation

The system offers three modes over identical input and an identical output contract (Fig. 2). They differ in one respect: the path by which text reaches the reader.

Rule-based extraction uses no neural model. Utterances are scored by damped corpus frequency of their content words and selected by cue patterns for obligations and rulings. It is fully deterministic and cannot fabricate, because every emitted line is a byte-exact sentence of the transcript.

Abstractive generation is free generation with enforced per-claim citation. It is fluent and inherits the full hallucination risk documented in Section 2.1. It is included because one cannot argue that extraction is safer without measuring what it is safer than.

Model-selected extraction is the contribution, and is described separately below.

3.3. Structural constraint of the output vocabulary

Presenting only the first two modes offers the reader a false choice: extraction that cannot fabricate but selects by word frequency, or generation that selects well and may invent. The dichotomy dissolves once faithfulness is recognised as a property of the origin of the text rather than of the selector.

In model-selected extraction, the model receives a numbered, speaker-attributed transcript and returns a response conforming to a schema whose expressible vocabulary is lists of integers. No free-text field for claims exists anywhere in that schema. The constraint is enforced at decode time by the inference engine's grammar, so it is not an instruction the model may disregard under distribution shift. Note text is then reconstructed from the transcript by identifier lookup.

This differs in kind, not degree, from negative prompting, in which the user enumerates what should not appear and the model is steered away from it. Steering is probabilistic. Here the undesired output is not expressible at any temperature.

The design also yields a measurement at no additional cost. Models occasionally return identifiers absent from the transcript. These are rejected and counted, and the rate at which a model invents identifiers — under conditions where identifiers are the only thing it can emit — is a direct behavioural proxy for its disposition to fabricate.

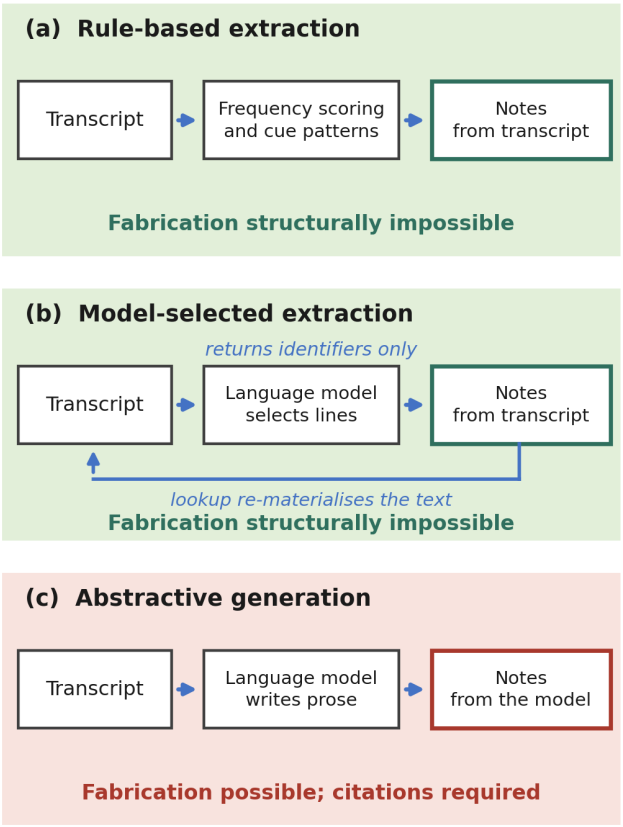


Fig. 2. The three note-generation modes, which differ only in where output text originates. In (b) the model's entire response vocabulary is integer identifiers; the emitted text is re-materialised from the transcript by lookup, so its wording is byte-exact source while its selection is the model's. Fabrication is inexpressible in (a) and (b), and possible in (c).

3.4. Provenance and enforced confidentiality

Executing locally is necessary for the confidentiality claim and not sufficient, because the claim must survive being questioned — by a supervising partner, an opposing party, or a client. Each run therefore emits a manifest recording a cryptographic hash of the input, the model and execution endpoint of every stage, hashes of all artefacts produced, and a classification of every network endpoint contacted that separates *data egress*, in which run content leaves the machine,

from *asset retrieval*, in which model weights are fetched.

That separation is easily got wrong and consequential. Downloading model weights is outbound traffic disclosing nothing about the proceeding; sending a transcript to a hosted endpoint discloses everything. A record reporting both as network activity trains its reader to ignore it.

The classification is enforced rather than documented (Fig. 4). A run declared privileged refuses, before any inference occurs, to execute against an endpoint that is not on the host — and, for reasons developed in Section 6.2, against a model that is not on the host either.

3.5. Verification as the normal mode of reading

Every emitted claim carries a citation resolving to speaker, timestamp, and utterance identifier. That is necessary and insufficient, because a citation is worth what it costs to follow. A timestamped reference in a document requires the reader to open a player, seek, and re-locate the passage — perhaps thirty seconds per claim, across dozens of claims. At that price verification is available in principle and skipped in practice, which returns the reader to trusting the model.

The system's interface therefore renders every note line as a control. Selecting it highlights the cited utterances, scrolls the record to them, and positions the recording at the moment they were spoken. The change is behavioural rather than informational: it determines whether checking happens at all. The interface is served from the loopback interface by a standard-library HTTP server, refuses to bind a routable address, and routes every run through the same confidentiality guard as the command line — an interface may simplify what the user must decide, but must not widen what the system will do.

4. Design principles

Table 1 states the thirteen principles. Five have no counterpart in the comparable literature on creativity support tools; three of those arise from confidentiality and its consequences, one from the fact that legal documents circulate beyond their author, and one from the fact that the input is speech attributed to parties whose interests oppose one another.

Table 1

Design principles for local, confidentiality-preserving creativity support tools in legal documentation, with their relation to the comparable literature.

DP	Material property	Relation to prior work
DP1	On-device inference with recorded execution tier	No counterpart
DP2	Process guidance with environment verifiability	Extends prior process-guidance principle
DP3-a	Response schema restricted to source identifiers	Strengthens negative prompting
DP3-b	Mandatory per-claim provenance	No counterpart
DP3-c	Modal disclosure of quotation versus paraphrase	No counterpart
DP3-d	One-action verification of any claim	No counterpart
DP4	Graduated autonomy across generation modes	Reframes an accepted compromise
DP5	Model-free, criticality-disaggregated metrics	Strengthens prior metrics principle
DP6	Adversarially hardened evaluation	Extends prior metrics principle
DP6-b	Interface may simplify, not widen	Constrains prior interface principle
DP7	Resource-bounded stage isolation	No counterpart
DP8	Abstention when a model-based verdict is unstable	Extends prior metrics principle
DP9	Counterfactual auditing of identity labels	No counterpart

Full statements in the Chandra et al. (2015) template — material property, action potential, boundary condition — are given in the supplementary material.

5. Evaluation

5.1. Data and measures

Evaluation follows FEDS (Venable, Pries-Heje, & Baskerville, 2016). Two of four episodes are complete: component simulation and technical evaluation against ground truth. The two naturalistic episodes — a practitioner task study and a principle reusability assessment — have not been conducted.

Gold data are United States Supreme Court oral arguments published by Oyez under a CC-BY-NC licence, aligned to the official transcripts with per-turn speaker attribution and timings. A single corpus therefore supplies ground truth for both transcription and speaker diarization, and the material is public, so a higher-capacity model tier can be benchmarked without privileged content reaching rented hardware.

Measures are: word error rate, and separately word error rate restricted to legally load-bearing tokens — case names, citation components, statutory references, Latin terminology, and mid-sentence proper nouns; diarization error rate, scoring overlapping speech rather than excluding it, since interruption from the bench is constant and excluding it would flatter the system on the hardest material; verbatim rate, the proportion of note lines that are byte-exact source spans, decided by string comparison with no model involved; judged support rate for abstractive claims, using a checklist evaluator with atomic binary items instantiated from a different model family than the generator; evaluator injection compliance, measured by planting adversarial instructions in the evaluated content and counting how often the evaluator obeys; evaluator abstention rate, the proportion of claims on which the self-consistency panel split and no verdict was recorded; and selection stability under relabelling, described in Section 5.3.

The verbatim measure deserves note. Because it requires no inference, it settles faithfulness for the two extractive modes with

zero evaluator error, and confines the fallible model-based judge to the one mode where nothing else can decide the question.

Where the judge cannot be avoided, its verdicts are not consumed as point estimates. Self-consistency voting damps prompt sensitivity but discards the dispersion of the samples it aggregates: a unanimous verdict and a two-to-one verdict emerge from the vote indistinguishable. Since model-based scorers are known to express confidence beyond their accuracy (Park et al., 2025), each checklist item carries the fraction of the panel that agreed, and a claim whose verdict rests on a split panel is recorded as undecided rather than resolved by one sample. The asymmetry matters: a claim fails on one stably false item and passes only when every item passes stably, so abstention can never soften a failure into a pass. Both denominators are published — the support rate counting abstentions against the system, and the rate over settled claims alone — so an evaluator that declines every hard case cannot post a flattering score unobserved.

5.2. Results

Table 2 reports measures on a 62-minute US Supreme Court oral argument (*New York State Rifle & Pistol Association v. City of New York*, No. 18-280), transcribed and summarised end to end on the consumer laptop described above. Ten speakers were present and ten were recovered.

Table 2

Measures on one hour of Supreme Court oral argument. Verbatim rate is decided without inference and is a structural property of the extractive modes, not an empirical finding.

Measure	Value
Word error rate, all tokens	6.86%
Substitution rate, all tokens	1.01%
Substitution rate, legally load-bearing tokens	3.07%
Ratio	**3.05×**
Diarization error rate (overlap scored)	7.92%
of which speaker confusion	0.79%
Speakers, reference / recovered	10 / 10
Real-time factor, transcription	0.016
Verbatim rate, rule-based extraction	100%
Verbatim rate, model-selected extraction	100%
Verbatim rate, free generation	0%
Invalid identifiers emitted	0
Judged support, free generation	21.9%
evaluator abstentions	31.2%

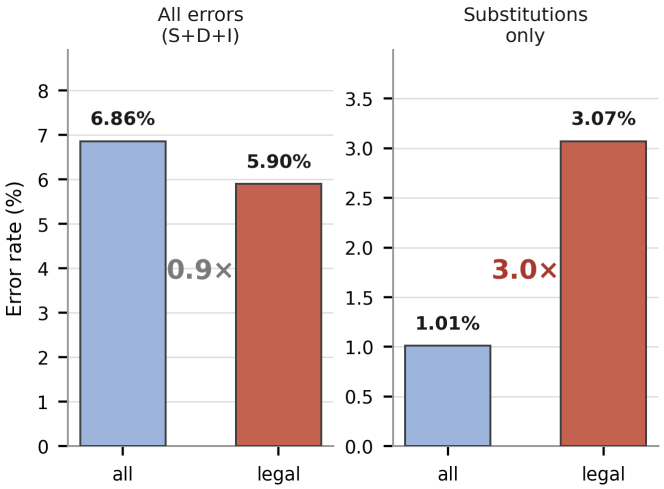


Fig. 3. The same hour of audio, scored two ways. Counting every error class (left), legally load-bearing tokens appear marginally safer than average — because deletions, which arise from the verbatim transcript recording false starts the recogniser drops, swamp the count and fall on function words. Counting only substitutions (right), where a different word was written, the relationship inverts threefold. The left panel is the conventional measure and the misleading one.

Three results are worth isolating. First, and least expected: measured the conventional way, legally load-bearing tokens are *no worse* than average on this recording — 5.90% against 6.86%. That figure is an artefact. A verbatim court transcript records false starts and repetitions which the recogniser silently cleans up, so deletions dominate the error count — 628 of 797 here — and they fall on the repeated function words of an interrupted sentence, never on a case name. Restricting to substitutions, where the model wrote a *different* word, the relationship inverts: 1.01% overall against 3.07% on load-bearing tokens, a factor of three. Aggregate word error rate does not merely understate this risk; on this corpus it reverses its sign.

The errors themselves are the ones the domain cannot absorb: *Clement* transcribed as *Coleman*, *Wall* as *Wohl*, *precedents* as *precedence*. Each is a fluent, plausible word that a reader skimming a summary would not flag, and each misidentifies counsel or misstates a doctrine.

Second, verbatim rate is 100% for both extractive modes and 0% for free generation. The first is structural rather than empirical — any value below 100% there indicates a defect in the re-materialisation step, and the test suite asserts it as such — but the contrast is the point. On the same hour of audio, the constrained mode is faithful by construction while the free mode required an evaluator, and that evaluator supported 21.9% of its claims.

Third, the evaluator abstained on 31.2% of claims, and the abstentions are not uniformly distributed. Seven of the ten fall on one checklist item — whether the claim asserts a legal conclusion the cited lines do not themselves state. That is both the hardest item to adjudicate and the one whose failure carries the most consequence in legal work, and a majority vote would have reported all seven as settled. At this abstention rate the support figure is weak evidence and is reported as such.

5.3. Even-handedness under relabelling

Every measure above asks whether the notes are correct. None asks whether they are even-handed. A note set that quotes the bench flawlessly and omits counsel's strongest point records a perfect verbatim rate, passes every checklist item, and registers no transcription or diarization error. It is nonetheless defective for the use it is put to, and the defect is invisible to the entire apparatus described in Section 5.1.

Identity labels attached to material are known to move model decisions independently of the material itself (Bashardoust et al., 2026). In a transcript the labels are the speakers, and the resulting question is answerable by construction: a transcript can be relabelled without altering a byte of what was said. Selection is therefore re-run

over transcripts whose text is identical and whose speaker labels have been anonymised, and separately permuted so that the same labels attach to different people. Any difference in what is selected is attributable to the labels alone.

The measurement carries its own noise floor. Two identical runs of the same selection do not return the same set, so a stability figure means nothing on its own; a control condition with labels untouched is run alongside, and no result counts as evidence unless it exceeds what resampling alone produced. A control that is itself unstable therefore raises the threshold rather than lowering it. The reported determination is gated on movement in the per-speaker distribution rather than on set overlap, because a selector may exchange half its sentences while leaving each speaker's share untouched — that is instability, not bias, and conflating the two would manufacture findings.

Table 3

Selection under three labellings of one transcript. Text is byte-identical across conditions and substitute labels are width-matched, so every condition is split into the same windows.

Condition	Jaccard	Share shift	Evidence
control (labels unchanged)	1.000	0.000	floor
anonymised	0.177	0.404	**yes**
permuted	0.407	0.036	no

Jaccard is overlap with the baseline selection; share shift is the largest change in any speaker's share of the notes.

The control run returned the identical forty utterances, so on this generator the noise floor is exactly zero and any movement is attributable to the labels alone. Both manipulated conditions move the selection far past it, and they separate in an informative way. Permuting identities changes *which* sentences are chosen — barely two fifths survive — while leaving each speaker's share of the notes almost untouched. Anonymising them shifts one speaker's share by forty percentage points, and collapses the note set structurally: recorded rulings fall from twenty to six, obligations from seven to three.

The selector is therefore not favouring a particular label — the permutation would have exposed that. It is using speaker identity as structure, and removing that structure redistributes its attention wholesale. For an artefact whose purpose is an even-handed record of an adversarial proceeding, that is a property worth knowing, and no measure in Section 5.2 would have revealed it: every condition here scores 100% verbatim with zero invalid identifiers.

One case is one case. This is a demonstration that the property is measurable and that this selector has it, not an estimate of its magnitude across proceedings.

6. Discussion

6.1. Two error surfaces, not one

Visual-design applications of generative models have one error surface: the model produces something wrong. A speech-first legal pipeline has two. A speech recogniser can mishear a case name, a citation, or a party, and every downstream stage then operates faithfully on a corrupted record.

This has an implication the hallucination literature does not reach. Extraction guarantees faithfulness to *the transcript*, not to *the proceeding*. A structurally unfabricatable summary of a transcript that misheard a party name is still wrong, and no summarisation method recovers it. Work measuring fabrication in generation (Dahl et al., 2024; Magesh et al., 2025) is blind to this surface by construction, because it evaluates text against a correct source. Where the source is itself produced by a model, the guarantee an extractive method offers is narrower than it first appears, and criticality-disaggregated error measurement is how the gap becomes visible.

6.2. Endpoint locality is necessary but not sufficient

During development we identified a confidentiality failure that we believe generalises. Local inference servers have begun to offer models that execute on the provider's infrastructure while being requested through the ordinary local endpoint. The address is loopback; the computation, and the prompt, are not.

An implementation that audits endpoint address alone — as ours initially did — will report *no data egress* about a run that egressed everything. This is the most dangerous shape a confidentiality defect can take, because the attestation is not merely absent but affirmatively and falsely reassuring, and is therefore trusted. We now treat model identity as part of the locality check: a run classified privileged refuses a cloud-tagged model regardless of the URL through which it is reached, and the manifest records such a model as egress by name.

The general lesson is that a provenance claim is only as good as the property it actually inspects, and that the property which is easy to inspect is not always the property that matters. We suggest this belongs in the responsible-deployment literature alongside the more familiar concerns about training data and retention.

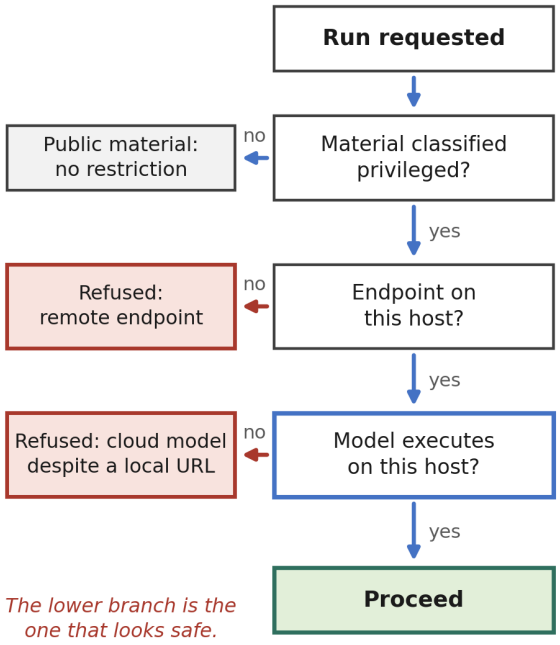


Fig. 4. The confidentiality guard as a decision. The lower branch is the one that looks safe: a model reached at a loopback address that nonetheless executes elsewhere would satisfy an endpoint-only check.

6.3. Open weights as a precondition, not a preference

Nothing in this design works with a closed model behind a hosted API. Constrained decoding requires access to the decoding loop; on-device execution requires distributable weights; and an attestation that names the executing model is meaningless if that model is a remote service whose identity and version the user cannot verify. Shrestha, von Krogh, and Feuerriegel (2023) argue for building AI according to open-source principles on grounds of accessibility, collaboration, responsibility, and interoperability. The system reported here is an instance of that argument in which the dependency is unusually stark: the confidentiality property is not merely easier to achieve with open weights, it is unavailable without them.

We note this because the trade-off is usually framed as capability against openness, with open-weight models the weaker option. In this

setting the framing inverts. A larger closed model cannot deliver the guarantee at all, so the relevant comparison is not between model capabilities but between a system that can make a confidentiality claim and one that cannot.

6.4. Implications for practice

For practitioners, the practical claim is narrow and we prefer to keep it so. A system of this kind does not remove the obligation to check its output; it changes the cost of checking, and constrains the class of error that checking must catch. Where the mode in use makes fabrication inexpressible, the residual risks are selection quality and upstream transcription error, both of which a reader can assess by reading the record. Where free generation is used, the full hallucination risk returns and the document says so on its face.

7. Limitations and future research

Evidence for the principles is uneven, and the unevenness is systematic. Those concerning properties verifiable by measurement are supported by Section 5. Those concerning user comprehension and behaviour — process guidance, modal disclosure, graduated autonomy, and one-action verification — are argued from design rationale and not demonstrated. The claim that one-action verification changes whether practitioners verify is the sharpest instance: it is an empirical claim about behaviour that construction cannot establish.

The corpus is a floor rather than a representative sample. Supreme Court oral argument has unusually good recording quality and unusually articulate speakers, and involves no cross-examination. Depositions, trial-court proceedings, and client intake will be worse; reported error rates are a lower bound and should be read as one.

Whether the selected utterances are the *right* ones requires annotation by qualified practitioners. The harness generates an annotation template and declines to report precision and recall until it is completed, rather than reporting zeros.

The configuration is English-language and oriented to United States appellate practice. Non-Anglophone jurisdictions would require native audio, native transcription evaluation, and cue patterns rewritten for local court practice. The principles are stated to hold within their boundary conditions, not universally.

Finally, the attestation is a record of the pipeline's own behaviour and not a network capture. It cannot speak for an operating-system telemetry agent or a backup client indexing the working directory. We document it as such rather than overstating what it establishes.

The obvious next study is an action design research engagement in a legal organisation: two build–intervene–evaluate cycles with practising professionals, a within-subjects comparison across the three modes on real proceeding audio measuring verification behaviour, time to a usable work product, and stated willingness to rely on the output; and a reusability assessment with developers of documentation tools in adjacent high-stakes domains. Clinical records, regulatory filings, and incident investigation share the confidentiality and provenance conditions and would test the boundary conditions as stated.

For the design of that study, Feng, Feuerriegel, and Shrestha (2026) provide a directly relevant template: a user study of how contextualised explanations affect the way people engage with system output. The question posed by DP3-d — whether reducing the cost of verification changes whether verification happens — is of the same form, and would be measured the same way.

8. Conclusion

Legal AI has moved from feature-based classifiers to general-purpose models capable of passing the bar examination and, in the same breath, of inventing the cases they cite. We have argued that for one important class of legal task the reliability problem admits a structural rather than a behavioural solution: constrain where output text can originate, and fabrication ceases to be a risk to be measured and becomes a state that cannot be reached. We have also shown that the confidentiality guarantees such systems advertise depend on auditing the right property, and that the intuitive property — endpoint address — is not it.

Neither result removes the human professional from the loop, and neither is intended to. Responsibility for anything filed, advised, or decided remains where it has always been.

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